

Session 6. FINAL PROJECT

(de)multiplexer based on cascaded Mach-Zehnder Interferometers (MZIs)

Introduction.

This project focuses on designing and simulating a CWDM (de)multiplexer based on cascaded Mach-Zehnder Interferometers (MZIs) on a silicon-on-insulator (SOI) platform. The students will:

- Simulate propagation modes in an SOI waveguide.
- Design directional couplers to obtain specific coupling constants.
- Develop a SAX model to analyze a basic MZI component.
- Extend the design to cascaded MZI structures.
- Create the layout of the basic BB, filter structure and the test structures.

References.

The project is based on the article: Tzu-Hsiang Yen and Yung-Jr Hung, "Fabrication-Tolerant CWDM (de)Multiplexer Based on Cascaded Mach-Zehnder Interferometers on Silicon-on-Insulator," Journal of Lightwave Technology, vol. 39, no. 1, 2021.

Part I. Design and Simulation of Individual Components

1. Simulating Propagation Modes in SOI Waveguides (2 hours)

- **Steps:**
 1. Define the SOI waveguide structure (450 nm width, 220 nm thickness).
 2. Simulate fundamental TE and TM modes.
 3. Extract effective index (n_{eff}) and group index (n_g) values.

2. Designing Directional Couplers (2 hours)

- **Objective:** Achieve coupling constants as shown in **Figure 5(b)**.
- **Steps:**
 1. Simulate directional couplers in GDSFactory.
 2. Sweep different gap values and coupling lengths.
 3. Extract coupling coefficients (e.g., 0.5, 0.3, 0.23, etc.).

3. Cascaded MZI Filter and SAX Model (2 hours)

- **Steps:**
 1. Extract the S-matrix from individual components.
 2. Define two arms of the MZI with different lengths.
 3. Implement the phase condition as per **Equation (1)**.
 4. Simulate transmission spectrum and confirm free spectral range (FSR).
 5. Create SAX model connecting multiple MZIs in series using the designed directional couplers
 6. Validate spectral alignment of outputs (Achieve transmission spectrum as shown in **Figure 6(a)**).

Part II. Filter and die layout

4. Layout of the CWDM (de)multiplexer filter employing GDSFactory+ (1.5 Hours)

Objective: Create the schematics for S1 and S2 and connect them to generate the layout of the CWDM (de)multiplexer filter employing GDSFactory+.

Steps:

1. Create the schematics for S1 and S2 and connect them to generate the layout of the CWDM (de)multiplexer filter employing GDSFactory+.
2. Define the layout of directional couplers, waveguides, and phase shifters as needed.
3. Ensure proper spacing between components to avoid unwanted coupling or interference.

5. Layout of the Whole Die and Test Structures (1 Hour)

Objective: Finalize the full die layout with test structures for fabrication.

Steps:

1. Integrate the CWDM (de)multiplexer layout with input/output connections.
2. Add test structures (e.g., individual directional couplers, waveguides, and MZIs) to enable characterization of the filter.
3. Export the final .GDS file for submission and verification.

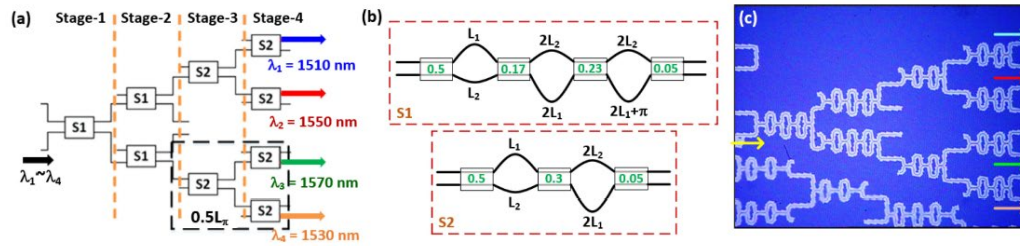


Fig. 5. (a) The schematic diagram of a 4-channel CWDM (De)MUX based on four-stage cascaded MZIs and (b) its layout of the MZI units for the S1 and S2 stages. (c) Top microscopic view of as-realized cascaded MZIs for CWDM (De)MUX.

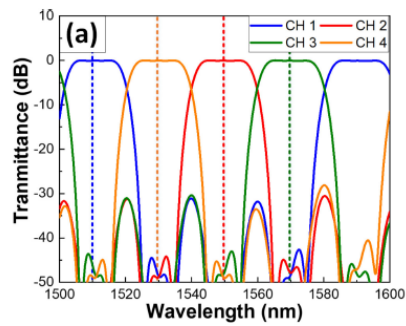


Figure 6(a)